

- 2 Biodiversity is important in maintaining the stability of an ecosystem. Biodiversity can be reduced by the introduction of new species to an ecosystem.

The red quinine tree, *Cinchona pubescens*, was introduced to the Galapagos Islands in the 1940s. By 2010, it had covered 110 000 hectares of Santa Cruz, one of the larger Galapagos islands.

- (a) A group of scientists studied the impact of the red quinine tree on the plant biodiversity of Santa Cruz.

Line transects were used to study an area of 32 hectares in the hills of Santa Cruz.

- (i) Suggest **three** variables that the scientists needed to standardise in this study.

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[3]

In addition to the line transects, the scientists set up 14 sample plots in the study area.

- (ii) Suggest a method that could be used for randomly selecting the position of the plots.

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[1]

- (b) The scientists decided to use Simpson's Index of Diversity to calculate the plant biodiversity of the study plots.

State what data they needed to collect to calculate Simpson's Index of Diversity.

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[1]

(c) Over a 7-year period the scientists measured the:

- ground covered by red quinine trees
- ground covered by other plant species
- ground **not** covered by plants.

Fig. 2.1 shows the results.

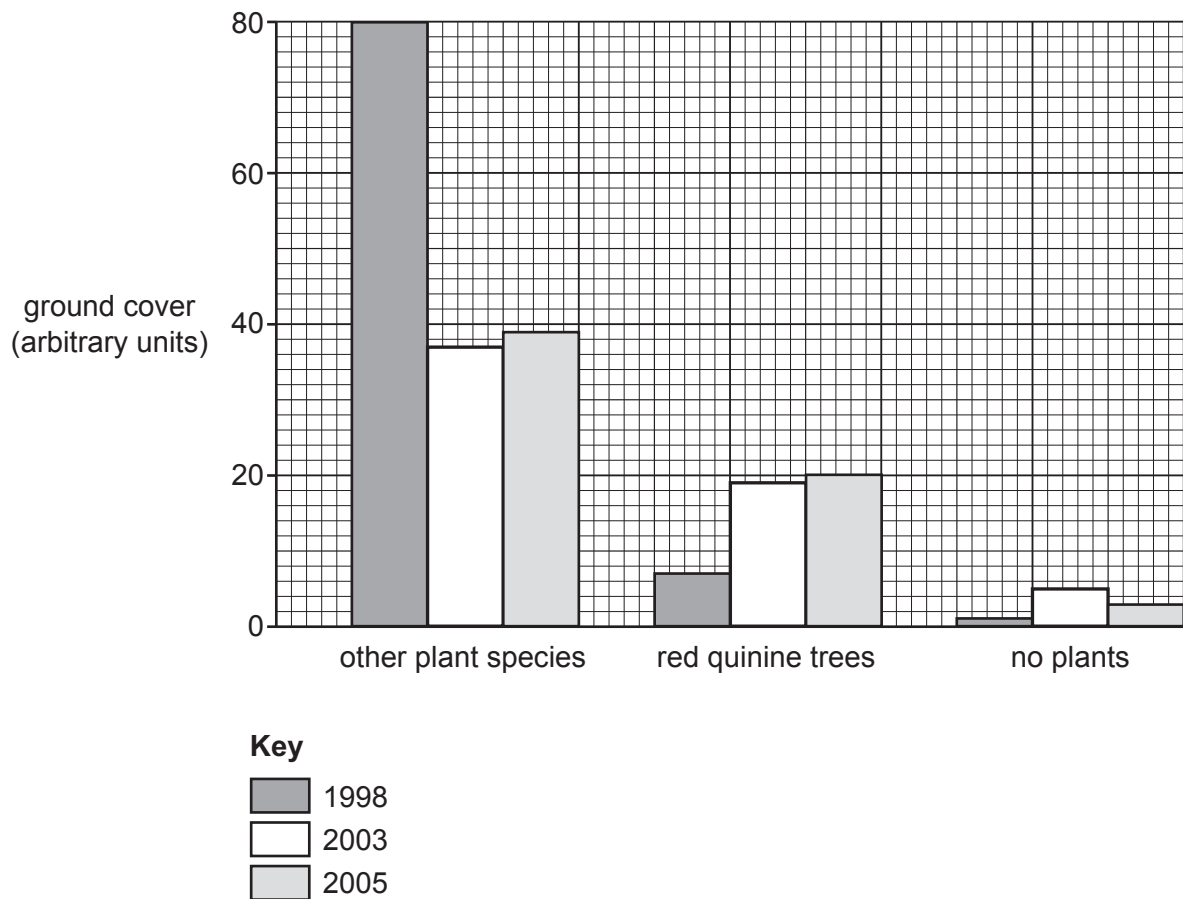


Fig. 2.1

Calculate the percentage change in the ground covered by red quinine trees in the plots from 1998 to 2005.

..... % [1]

(d) The scientists suggested the hypothesis:

The presence of red quinine trees in the hills of Santa Cruz causes a decrease in biodiversity.

The scientists carried out statistical tests on their studies of species diversity.

The probability values (p) from the results of the statistical tests are shown in Table 2.1.

Table 2.1

	value of p
decrease in species diversity 1998–2003	< 0.001
decrease in species diversity 2003–2005	> 0.05

Evaluate the data in Fig. 2.1 and Table 2.1 and discuss the extent to which the data supports or does **not** support this hypothesis.

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..... [4]

[Total:10]